Lexington Whalen

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Profile

Machine learning researcher specializing in efficient training and inference of large language models, with publications at EMNLP, ICML, CVPR, and ACL and an NSF Graduate Research Fellowship. Currently co-lead pretraining and inference optimization for SoftBank SB Intuitions' Sarashina LLMs, Japan's largest series of large language models; previously co-led NVIDIA's diffusion language models and led research teams in the U.S. and Japan.

Education

Georgia Institute of Technology — **Atlanta, GA**
M.S. in Computer Science, Specialization in Artificial Intelligence — *Expected Dec. 2026*
NSF Graduate Research Fellow.

University of South Carolina, Honors College — **Columbia, SC**
Accelerated B.S./M.S. in Computer Science — *Jan. 2021 – Aug. 2024*
Outstanding Senior, Class of 2023.

Preprints & Technical Reports

[1] **L. Whalen**, Y. Ito, R. Sakamoto. "KBBQ: A Predictive Noise Law and the Limits of Spectrum Flattening in FP4 Quantization." Technical Report, 2026. [SB Intuitions]

[2] **L. Whalen**, Y. Ito, R. Sakamoto. "Teaching Diffusion to Speculate Left-to-Right." Technical Report, 2026. Under review at TMLR. [SB Intuitions]

Peer-Reviewed Publications

[1] Y. Fu, **L. Whalen**, A. Garg, C. Wu, M. Khadkevich, N. Oswald, E. Xie, D. Egert, S. Turuvekere Sreenivas, S. Diao, C. Yu, Y. Yu, W. Chen, S. Norouzi, S. Lan, L. Zhu, J. Wang, J. Jiang, M. Mardani, M. Maghoumi, S. Han, A. Jukić, N. Tajbakhsh, J. Kautz, P. Molchanov. "Nemotron-Labs-Diffusion: A Tri-Mode Language Model Unifying Autoregressive, Diffusion, and Self-Speculation Decoding." **EMNLP**, 2026. [NVIDIA]

[2] Y. Fu*, **L. Whalen***, Z. Ye, X. Dong, S. Diao, J. Liu, C. Wu, H. Zhang, E. Xie, S. Han, M. Khadkevich, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Efficient-DLM: From Autoregressive to Diffusion Language Models, and Beyond in Speed." **ICML**, 2026. [NVIDIA]

[3] C. Shih, C. Li, C. Yu, H. Fang, S. Li, W. Hsin, **L. Whalen**, H. Suh, G. Eisenhauer, L. Liu, Y. (Celine) Lin. "NeRArch-Sim: A Unified Simulator for Benchmarking and DSE of Neural Rendering Accelerators." **ISCA**, 2026. [Georgia Tech]

[4] **L. Whalen***, Z. Du*, H. You*, C. Li, S. Li, Y. (Celine) Lin. "Early-Bird Diffusion: Investigating and Leveraging Timestep-Aware Early-Bird Tickets in Diffusion Models for Efficient Training." **CVPR**, 2025. [Georgia Tech]

[5] Z. Ye, Z. Wang, K. Xia, J. Hong, L. Li, **L. Whalen**, C. Wan, Y. Fu, Y. (Celine) Lin, S. Kundu. "LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering." **ACL**, 2025. [Georgia Tech]

[6] A. Murphy, **L. Whalen**, S. Dubinsky, M. Gavin, J. F. Bailyn, J. Ginn. "On 'historical unity' of Russian and Ukrainian: A linguistic perspective on language conflict and change." **LSA**, Apr. 2023. [University of South Carolina]

* Equal contribution

Patents

[1] **L. Whalen**, Y. Ito, R. Sakamoto. 「拡散ベースの投機的ドラフトモデルにおける受理長改善アルゴリズム」 Japanese Patent (to be filed), 2026. [SB Intuitions]
"Algorithm to Improve Acceptance Length of Diffusion-Based Speculative Draft Models."

[2] Y. Fu, **L. Whalen**, X. Dong, S. Diao, C. Wu, E. Xie, S. Han, J. Kautz, Y. (Celine) Lin, P. Molchanov. "Training Framework for Converting Autoregressive Language Models into Faster Diffusion Language Models." U.S. Patent (to be filed), 2026. [NVIDIA]

Research Experience

SoftBank – SB Intuitions **Tokyo, Japan**
Pretraining & Inference Optimization Research Lead *Apr. 2026 –*

- Co-lead pretraining and post-training optimization research for the Sarashina series, Japan's largest LLMs, at SoftBank's SB Intuitions R&D Division.
- Lead novel quantization efforts, spanning models up to hundreds of billions of parameters.
- Built SB Intuitions' first speculative-decoding inference stack for the Sarashina family; introduced training-time drafter-alignment methods that raise the accepted draft length by 21–76% over a position-uniform baseline.
- Serve as a key technical voice in Japanese-language cross-team discussions on pretraining, post-training, and inference optimization, bridging research and engineering.
- Collaborate with leading Japanese and international technology companies and government agencies to evaluate GPU resources and co-develop long-term compute strategies for large-scale training.

NVIDIA **Atlanta, GA & Santa Clara, CA**
Efficient Deep Learning Intern & Data Filtering Challenge Lead *Dec. 2024 – Mar. 2026*
Mentors: Dr. Pavlo Molchanov & Yonggan Fu (NVIDIA Research).

- Co-developed Nemotron-Labs-Diffusion (EMNLP 2026), a tri-mode (autoregressive / diffusion / self-speculation) language model family at 3B–14B scale; the 8B model decodes 6× more tokens per forward pass than Qwen3-8B at comparable accuracy (up to 4× higher throughput); released on Hugging Face with ∼1M total downloads.
- Co-first author of Efficient-DLM (ICML 2026), converting pretrained autoregressive models into diffusion LMs; the 8B model reaches +5.4%/+2.7% higher accuracy at 4.5×/2.7× higher throughput than Dream 7B / Qwen3 4B (open-weights on Hugging Face); presented directly to CEO Jensen Huang.
- Built the training, evaluation, and inference infrastructure adopted by 5+ internal research teams for vision-language models, embeddings, and latent reasoning.
- Organized NVIDIA's 2025 International Data Filtering Challenge for Training Edge Language Models across 30+ university and industry teams.

Georgia Institute of Technology **Atlanta, GA**
Graduate Researcher *Aug. 2024 – Dec. 2025*

- Led a team of 3 Ph.D. and Masters students (co-first author) to cut diffusion-model training time by 2.9–5.8× (up to 10.3× vs. standard train-prune-finetune) while maintaining generation quality; CVPR 2025.
- Algorithm lead for the United States' ARPA-H THOR healthcare initiative: reduced cancer detection sensor degradation by over 70%, extending sensor lifetime from a few hours to several days.
- Engineered resource-efficient ML architectures powered by energy from the human body, collaborating with teams from Stanford, Northwestern, Carnegie Mellon, and MIT.

Honors & Awards

- SoftBank Group Representative, 2026 Entrance Ceremony — Mar. 2026
- NSF Graduate Research Fellowship ($150,000 value) — Apr. 2024
- Toyo University Exchange Student Representative — Jul. 2023
- University of South Carolina Outstanding Senior, Class of 2023 — Feb. 2023
- U.S. Department of State Critical Language Scholar of Japanese — Jan. 2023
- University of South Carolina Top Scholar ($100,000+ value) — Aug. 2020

Invited Talks

- "拡散言語モデル：基本から応用まで", SoftBank – SB Intuitions — Jul. 2026
  Diffusion Language Models: From Fundamentals to Applications
- 入社式スピーチ, SoftBank — Apr. 2026
  Entrance Ceremony Speech

Service & Outreach

- Serve as an informal advisor on U.S.–Japan AI and technology policy, engaging with U.S. Embassy Tokyo and Japanese government stakeholders.

Language Skills

English (Native) · **Japanese** (Native Level, JLPT N1 Certified) · **Mandarin** (Intermediate)